

Minimal Quantum Logic: a brief comparison with Classical Logic

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Abstract

Semantically, Classical Logic (CL) is associated with Boolean algebras, while Minimal Quantum Logic (MQL) is associated with ortholattices, which are lattices where, notably, the distributive laws are not necessarily true. Deductively, when employing the sequent calculus, one has to impose restrictions on the inferences rules of QML, notably on the cut rule, in order to prove fewer theorems. Examples are given below.

Semantical Context and the failure of distributive laws

An **ortholattice** is a structure $\mathcal{B} = (B, \sqsubseteq, ', 1, 0)$ where $(B, \sqsubseteq)$ is a bounded lattice, with *maximum* **1** and *minimum* **0** and the unary operation $'$ (*orthocomplement*) satisfies the following conditions: $a'' = a$; if $a \sqsubseteq b$, then $b' \sqsubseteq a'$; and $a \sqcap a' = 0$. [1]

Some examples of ortholattices are:



Figure 1: 2

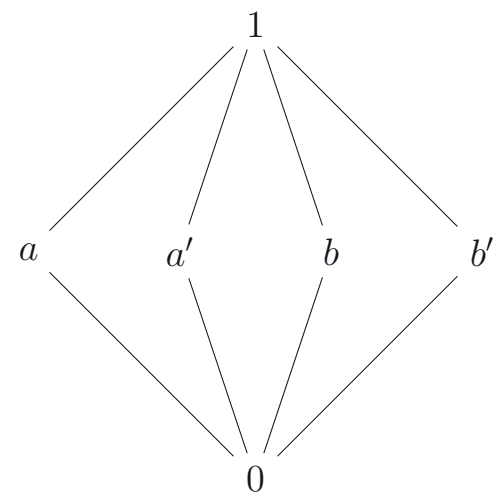


Figure 2: M_4 .

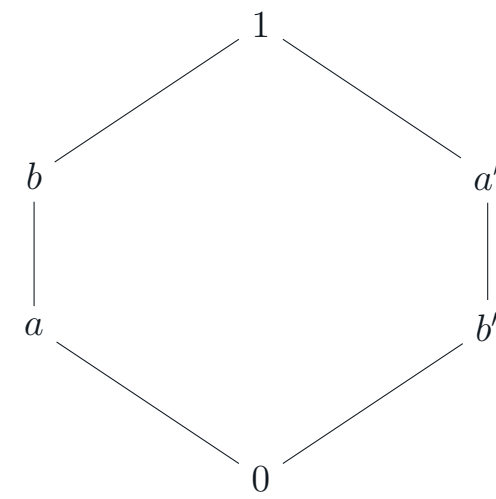


Figure 3: Hexagon.

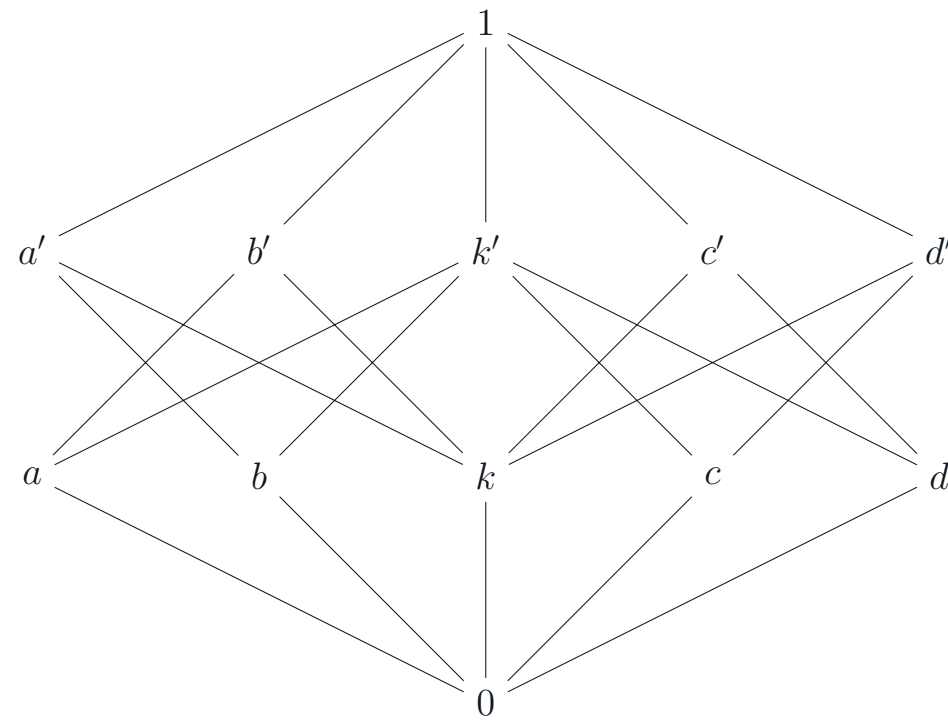


Figure 4: Another example of an Ortholattice.

In opposition to Boolean Algebras, in general, distributive laws don't hold for ortholattices. One can easily verify, on the previous example, that:

$$a \sqcap (b \sqcup c) = a \sqcap k' = a, \text{ but } (a \sqcap b) \sqcup (a \sqcap c) = 0 \sqcup 0 = 0$$

and

$$a \sqcup (b \sqcap c) = a \sqcup 0 = a, \text{ but } (a \sqcup b) \sqcap (a \sqcup c) = k' \sqcap k' = k'.$$

A formula A is a **logical truth** of Classical Logic (CL), also called **tautology**, if for any boolean algebra B , for any valuation v in B , $v(A) = 1$, where 1 is the maximum element.

Analogously, a formula A is a **logical truth** of Minimal Quantum Logic (MQL) if for any ortholattice B , for any valuation v in B , $v(A) = 1$, where 1 is the maximum element.

Deductive Systems and the cut rule restrictions

A **sequent** is an expression of the form $\Gamma \vdash \Delta$, where Γ, Δ are finite (possibly empty) sets of formulas.

An **inference rule** $\frac{\Gamma_1 \vdash A_1, \dots, \Gamma_n \vdash A_n}{\Gamma \vdash A}$ can be read as: if A_1 can be inferred from $\Gamma_1, \dots, A_n$ can be inferred from Γ_n , then A can be inferred from Γ .

A **deductive system** is composed of a set of inference rules which chained together give rise to formal proofs, also called derivations.

Let's introduce two distinct deductive systems, one for **CL**, adapted from [3], the other for **MQL**, adapted from [2] and [4], which are defined through the following inference rules:

Sequent Calculus for CL

$$\begin{array}{l} \frac{}{A \vdash A} \text{id} \\ \frac{\Gamma \vdash \Theta}{A, \Gamma \vdash \Theta} \text{LW} \\ \frac{\Gamma \vdash \Theta, A}{\neg A, \Gamma \vdash \Theta} \text{L}\neg \\ \frac{A_i, \Gamma \vdash \Theta}{A_1 \wedge A_2, \Gamma \vdash \Theta} \text{L}\wedge, \text{ where } i \in \{1, 2\} \\ \frac{A_1, \Gamma \vdash \Theta \quad A_2, \Gamma \vdash \Theta}{A_1 \vee A_2, \Gamma \vdash \Theta} \text{LV} \\ \frac{\Gamma \vdash \Theta, A \quad A, \Delta \vdash \Lambda}{\Gamma, \Delta \vdash \Theta, \Lambda} \text{Cut} \\ \frac{}{\perp \vdash A} \text{L}\perp \\ \frac{\Gamma \vdash \Theta}{\Gamma \vdash \Theta, A} \text{RW} \\ \frac{A, \Gamma \vdash \Theta}{\Gamma \vdash \Theta, \neg A} \text{R}\neg \\ \frac{\Gamma \vdash \Theta, A_1 \quad \Gamma \vdash \Theta, A_2}{\Gamma \vdash \Theta, A_1 \wedge A_2} \text{R}\wedge \\ \frac{\Gamma \vdash \Theta, A_i}{\Gamma \vdash \Theta, A_1 \vee A_2} \text{R}\vee, \text{ where } i \in \{1, 2\} \end{array}$$

Sequent Calculus for MQL

$$\begin{array}{l} \frac{}{A \vdash A} \text{id} \\ \frac{\Gamma \vdash \Theta}{\neg \Theta, \Gamma \vdash \neg} \text{L}\neg \\ \frac{A, \Gamma \vdash \Theta}{\neg \neg A, \Gamma \vdash \Theta} \text{L}\neg\neg \\ \frac{A_i, \Gamma \vdash \Theta}{A_1 \wedge A_2, \Gamma \vdash \Theta} \text{L}\wedge, \text{ where } i \in \{1, 2\} \\ \frac{A_1 \vdash \Theta \quad A_2 \vdash \Theta}{A_1 \vee A_2 \vdash \Theta} \text{LV} \\ \frac{\Gamma \vdash \Theta, A \quad A \vdash \Lambda}{\Gamma \vdash \Theta, \Lambda} \text{Cut-1} \\ \frac{\Gamma \vdash \Theta}{\Delta, \Gamma \vdash \Theta, \Lambda} \text{W} \\ \frac{\Gamma \vdash \Theta}{\Gamma \vdash \Theta, \neg \Gamma} \text{R}\neg \\ \frac{\Gamma \vdash \Theta, A}{\Gamma \vdash \Theta, \neg \neg A} \text{R}\neg\neg \\ \frac{\Gamma \vdash A_1 \quad \Gamma \vdash A_2}{\Gamma \vdash A_1 \wedge A_2} \text{R}\wedge \\ \frac{\Gamma \vdash \Theta, A_i}{\Gamma \vdash \Theta, A_1 \vee A_2} \text{R}\vee, \text{ where } i \in \{1, 2\} \\ \frac{\Gamma \vdash A \quad A, \Delta \vdash \Lambda}{\Gamma, \Delta \vdash \Lambda} \text{Cut-2} \end{array}$$

Note that if Cut-1 and Cut-2 were replaced by the Cut rule, the Sequent Calculus for MQL would be equivalent to the Sequent Calculus for CL. [2]

Furthermore, the MQL system has some restricted inference rules, namely the rules LV, R $\wedge$, L $\neg$, R $\neg$ and the cut rules. Following this, an interesting question may naturally arise:

What are the consequences of these restrictions?

First, the semantical difference between boolean algebras and ortholattices is matched by the difference between the sequent calculus for CL and the sequent calculus for MQL, because of the soundness and completeness meta-theorems enjoyed by both logics.

Second, the gap between the deductive systems lead to situations where it is substantially more complex to derivate certain formulas in MQL, due to the imposed restrictions on the inference rules.

Finally, there are some cases where it is even impossible to derivate a classical tautology in MQL. Some illustrative examples are presented below.

Classical and Quantum Theorems whose derivations are substantially more complex in MQL

Let A and B be formulas. An example of a **CL theorem** which also is a **MQL theorem** is:

$$(A \vee B) \vee (\neg A \wedge \neg B).$$

A CL derivation is:

$$\frac{\frac{\frac{}{A \vdash A} \text{id}}{A \vdash A, B} \text{RW} \quad \frac{\frac{}{B \vdash B} \text{id}}{B \vdash A, B} \text{RW}}{\vdash A, B, \neg A} \text{R}\neg \quad \frac{\frac{}{B \vdash B} \text{id}}{B \vdash A, B} \text{RW}}{\vdash A, B, \neg A \wedge \neg B} \text{R}\wedge \quad \frac{\vdash A, B, \neg A \wedge \neg B}{\vdash A \vee B, \neg A \wedge \neg B} \text{R}_1 \vee, \text{R}_2 \vee \quad \frac{\vdash A \vee B, \neg A \wedge \neg B}{\vdash (A \vee B) \vee (\neg A \wedge \neg B)} \text{R}_1 \vee, \text{R}_2 \vee$$

The inference rule $\text{R}\wedge$ in the previous derivation is not possible in MQL. Given that, a MQL derivation, which includes the admissible inference rule from a sequent to its contraposition (C^*), has the form:

$$\frac{\frac{\frac{}{A \vdash A} \text{id}}{A \vdash A, B} \text{W} \quad \frac{\frac{}{B \vdash B} \text{id}}{B \vdash A, B} \text{W}}{A \vdash A \vee B} \text{R}_1 \vee, \text{R}_2 \vee \quad \frac{\frac{}{B \vdash B} \text{id}}{B \vdash A, B} \text{W}}{B \vdash A \vee B} \text{R}_1 \vee, \text{R}_2 \vee \quad \frac{\vdash A, B, \neg A \wedge \neg B}{\vdash A \vee B, \neg A \wedge \neg B} \text{R}_1 \vee, \text{R}_2 \vee \quad \frac{\vdash A \vee B, \neg A \wedge \neg B}{\vdash (A \vee B) \vee (\neg A \wedge \neg B)} \text{R}_1 \vee, \text{R}_2 \vee$$

Having the admissible rule C^* means that if there is a derivation $\mathcal{D}$ of a sequent $\Gamma \vdash \Theta$, then there will be a derivation $\mathcal{D}'$ of $\Theta' \vdash \Gamma'$. $\mathcal{D}'$ may result from an extensive transformation of $\mathcal{D}$, which may increase its size.

Classical Theorems which are not MQL theorems

The formula:

$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$

is a **CL Theorem**, since it has a derivation in the Sequent Calculus for CL:

$$\frac{\frac{\frac{}{A \vdash A} \text{id}}{A, B \vdash A, A \wedge C} \text{LW, RW} \quad \frac{\frac{}{B \vdash B} \text{id}}{A, B \vdash B, A \wedge C} \text{LW, RW} \quad \frac{\frac{}{A \vdash A} \text{id}}{A, C \vdash A \wedge B, A} \text{LW, RW} \quad \frac{\frac{}{C \vdash C} \text{id}}{A, C \vdash A \wedge B, C} \text{LW, RW}}{\frac{A, B \vdash A \wedge B, A \wedge C}{A, C \vdash A \wedge B, A \wedge C} \text{LV} \quad \frac{A, B \vee C \vdash A \wedge B, A \wedge C}{A, B \vee C \vdash (A \wedge B) \vee (A \wedge C)} \text{R}_1 \vee, \text{R}_2 \vee \quad \frac{A \wedge (B \vee C) \vdash (A \wedge B) \vee (A \wedge C)}{\vdash \neg(A \wedge (B \vee C)), (A \wedge B) \vee (A \wedge C)} \text{L}_1 \wedge, \text{L}_2 \wedge \quad \frac{\vdash \neg(A \wedge (B \vee C)), (A \wedge B) \vee (A \wedge C)}{\vdash \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))} \text{R}_1 \vee, \text{R}_2 \vee$$

On the other hand, D is **not a MQL logical truth**. Consider that $v(A)=a$, $v(B)=b$ and $v(C)=c$ in the ortholattice of Figure 4. Then, $v(D) = a' \neq 1$. Hence, by the soundness meta-theorem, D is not a theorem of MQL.

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